Case A

Mrs. Gupta — 50-Year-Old Graphic Designer | Working Diagnosis: Cervical Postural Dysfunction / Upper Crossed Syndrome

Musculoskeletal Physiotherapy

■ SUBJECTIVE PROBLEM LIST

- Physical Impairments
- Dull, persistent cervical and suboccipital pain (1 year)
- Reduced end-range cervical extension
- Weakness of deep cervical flexors and lower scapular stabilisers (presumed)
- Difficulty sustaining prolonged screen-based work
- Screen-worker may have developed a belief that movement worsens the condition — address early to prevent deconditioning.
- Yellow Flags to Explore

■■ RED FLAGS

- Clinical Significance
- Action if Present
- Bilateral upper limb paraesthesia / hand clumsiness / gait disturbance
- Suggests cervical myelopathy — cord compression requires urgent imaging (Hoffmann's, Lhermitte's sign). Absent here.
- Urgent referral to spinal surgeon / emergency department. Do NOT mobilise.
- Unexplained weight loss, night sweats, night pain unrelieved by position change
- Possible malignancy — metastatic disease to cervical spine can mimic mechanical neck pain.
- Urgent GP referral for systemic investigation; do not proceed with manual therapy.

■ PATHOPHYSIOLOGY (KEY POINTS)

- This creates a "crossed" pattern of tightness and weakness across the shoulder girdle and cervical spine.
- The protracted head posture shifts the centre of gravity of the head (approx.
- 5 kg) anterior to the cervical spine, increasing compressive forces and erector muscle demand by approximately 4–5x per inch of forward displacement.
- Dull, heavy pain: Ischaemic muscle pain from sustained upper trapezius/levator scapulae loading — not sharp or electric because nerve roots are not compressed.
- Suboccipital heaviness: Sustained shortening and compression of the suboccipital extensors maintaining forward-head posture — sensitisation of local nociceptors.
- Worsening at workstation: Sustained static posture maintains the mechanical deformation — no opportunity for muscle recovery or joint decompression.

■ DIFFERENTIAL DIAGNOSES

1. Radiculopathy vs. Postural Dysfunction:

2. Spondylosis vs. Postural Dysfunction:

3. Inflammatory arthropathy (e.g., RA):

■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

Key Observation Findings

Posture: Protracted head posture (ear in front of acromion in sagittal plane), rounded/elevated shoulders, increased cervicothoracic junction kyphosis, reduced thoracic extension. Movement: Cervical rotation appears within normal limits; end-range extension reproduces the suboccipital heaviness and posterior neck discomfort — confirming the symptomatic movement direction. Flexion is full and pain-free. Soft tissues:

TEST 1

Cranio-Cervical Flexion Test (CCFT) What it Measures Endurance and segmental activation of the deep cervical flexors (DCF) — longus colli and longus capitis — which are the key stabilisers inhibited in Upper Crossed Syndrome. How to Perform 1. Patient lies supine with a small pillow to maintain neutral cervical position. 2. A pressure biofeedback unit (or palpation) is placed behind the neck, inflated to 20 mmHg baseline. 3. Patient is asked to perform a gentle nodding action ("tuck your chin toward your chest without lifting your head"), progressively increasing through 22, 24, 26, 28, 30 mmHg pressure targets. 4. Each level is held for 10 seconds; observe for substitution (SCM/scalene activation, jaw clenching, breath-holding).

TEST 2

Spurling's Test (Foraminal Compression Test) What it Measures Cervical nerve root compression — narrows the intervertebral foramen to reproduce radicular symptoms. How to Perform 1. Patient seated. 2. Passively laterally flex and rotate the cervical spine toward the symptomatic side. 3. Apply gentle axial compression (approximately 7 kg) through the top of the head. 4. Maintain for 10 seconds or until symptoms reproduced. 5. Test both sides and compare.

Outcome Measures

Neck Disability Index (NDI)

■ TREATMENT PLAN

- 5. Have you had any previous neck injuries, whiplash, or episodes like this before? And are you currently taking any medications?

• 6. How has the pain affected your daily life, sleep, and work?
• Exercise 1: Deep Cervical Flexor (DCF) Activation — Chin Tucks (Craniocervical Flexion)
• Supine lying, small towel roll under neck for neutral. Gently retract the chin ("make a double chin") without lifting the head. Hold 10 sec, 10 reps, twice daily.
• Perform standing against a wall — wall provides tactile feedback, reduces gravitational demand. Start with 5-second holds.
• Add 2.5 cm head lift (endurance challenge). Then progress to sitting, standing without support. Add resistance band at forehead for advanced loading.
• Exercise 2: Scapular Retraction and Depression (Lower Trapezius Activation)
• Seated or standing. Arms by sides. Gently squeeze shoulder blades together and DOWN (not up). Hold 5–10 sec, 10 reps, avoid shoulder shrugging. Three sets, twice daily.
• Prone lying ("Y" exercise): arms overhead in Y shape, gently lift arms off table squeezing shoulder blades together. Gravity assists initially.
• Seated cable row or resistance band row emphasising scapular retraction. Progress to single-arm dumbbell row at low load. Monitor neck pain — stop if upper trapezius substitutes.
• Manual Therapy: Soft Tissue Mobilisation (STM) to Upper Trapezius & Levator Scapulae
• Precautions (IMPORTANT — Osteoporosis):